

Soumya Prabha Maiti

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EDUCATION

Jadavpur University <i>B.E. in Electronics and Tele-Communication Engineering; CGPA: 9.66/10</i>	Kolkata, India Aug 2019 – Jun 2023 (Expected)
Nava Nalanda High School <i>Higher Secondary (WBCHSE); Percentage: 95%; WBJEE 2019 rank: 186</i>	Kolkata, India 2017 – 2019
Nava Nalanda High School <i>Secondary (WBBSE); Percentage: 95.29%</i>	Kolkata, India 2006 – 2017

EXPERIENCE

ADES Lab, Jadavpur University <i>Undergraduate Academic Researcher, Guide: Prof. Mrinal Kanti Naskar</i>	Kolkata, India Aug 2022 – Present
- Developed a novel lossy image compression algorithm and implemented using Python and OpenCV. - Documented the findings, showcasing the superior performance of the novel compression method over JPEG. - Explored other image processing applications like edge detection, denoising, etc.	
Texas Instruments India <i>Intern</i>	Bangalore, India Jun 2022 – Jul 2022
- Contributed to the development of a Python framework for automatic testing and validation of RF chips. - Reverse-engineered and understood an existing undocumented Lua codebase, to develop the Python framework. - Integrated measuring instruments (like voltmeters, oscilloscopes), current sources etc., using the PyVISA library. - Implemented a mechanism within the framework to detect and throw exceptions when the instruments were connected incorrectly, ensuring accurate and reliable testing. - Provided documentation to other team members on how to effectively use and maintain the Python framework.	

PROJECTS

Blog website <i>Flask, SQLite, HTML, CSS and Bootstrap</i> GitHub
- Implemented user authentication and password reset with JWT. - Designed intuitive user interfaces for creating, editing, and deleting posts. - Implemented pagination on the home feed, allowing for efficient loading and navigation of blog posts.
Tomato leaf disease classifier <i>TensorFlow, Flask, HTML, CSS and Bootstrap</i> GitHub Demo
- Trained a Convolutional Neural Network (CNN) model on a Kaggle dataset using TensorFlow. - Implemented image preprocessing techniques like data augmentation, to enhance the model's robustness. - Deployed to Hugging Face Spaces using Flask and Docker.

SKILLS

Programming Languages: C, C++, Python, MATLAB
Machine Learning: NumPy, Matplotlib, Pandas, Scikit-learn, TensorFlow, OpenCV
Web Development: HTML, CSS, JavaScript, Flask
Tools: Git, Linux, Windows, Docker, VS Code, LaTeX
Languages: Bengali (Native), English (Professional), Hindi (Elementary)

ACHIEVEMENTS

Engineering Faculty topper in 2nd year: Received Late Supriya Basu's Scholarship for 2021.
400+ problems solved on Leetcode, 150+ problems on GeeksforGeeks.

CERTIFICATES

Intro to Machine Learning by <i>Kaggle</i> View Certificate
Neural Networks and Deep Learning by <i>DeepLearning.AI on Coursera</i> View Certificate
Convolutional Neural Networks by <i>DeepLearning.AI on Coursera</i> View Certificate